

Additional Formatting Commands in Excel

In Microsoft Excel, **formatting** means changing the appearance of data in a worksheet to make it **easy to read, attractive, and professional**. Apart from the basic formatting like font size, bold, italic, underline, and borders, Excel also provides many **additional formatting commands**. These commands help to highlight important data and make reports more understandable.

1. Number Formatting

- Excel allows you to display numbers in different ways without changing the actual value.
- Types:
 - **Currency/Accounting** → Shows numbers with currency symbols like ₹, \$, £.
 - **Percentage** → Converts numbers into percentage format (e.g., 0.5 → 50%).
 - **Date & Time** → Shows data as date (12-09-2025) or time (10:30 AM).
 - **Decimal Control** → Increase or decrease decimal places.

Example:

If you type 1234.5 and apply currency format, it becomes **₹1,234.50**.

2. Conditional Formatting

- This feature highlights cells automatically based on conditions.
- Example: If marks are **less than 35**, the cell can turn **red**.
- Useful for:
 - Highlighting top scores in a class.
 - Marking overdue payments.
 - Tracking sales performance.

Example:

If a student has 90 marks, Excel highlights the cell **green** if the condition is Marks > 80.

3. Alignment Options

- **Text Alignment:** Left, center, right alignment inside cells.
- **Vertical Alignment:** Top, middle, bottom.
- **Text Orientation:** Rotate text diagonally or vertically.
- **Wrap Text:** Adjusts long text within the same cell.
- **Merge & Center:** Combines multiple cells into one and centers the text.

Example:

Merging cells A1:D1 and writing “**Annual Report**” in the center.

4. Cell Styles & Themes

- **Cell Styles:** Predefined formatting sets (font, color, borders) to quickly format cells.
- **Themes:** Change the overall appearance of the workbook (colors, fonts, effects).

Example:

Using a “Heading” cell style for all titles in a table.

5. Borders & Shading

- Borders help to **separate data** clearly.
- Shading/Fill color is used to **highlight specific cells**.

Example:

Drawing a border around a marksheet and shading column headers with **yellow color**.

6. Custom Formatting

- Allows users to create **their own display format**.
- Example: Showing phone numbers as (###) ###-#### or adding text like Marks: 85.

Drawing Toolbar in Excel

What it is:

The Drawing tools let you **insert and format shapes, lines, text boxes, WordArt, and effects** to create flowcharts, labels, diagrams, and annotated reports right on your worksheet.

Where to find it (modern Excel):

Insert → **Illustrations** → **Shapes** (then the **Shape Format** tab appears when a shape is selected).

Where it was (older Excel 2003):

A floating **Drawing** toolbar at the bottom with icons like **AutoShapes, Line, Arrow, Rectangle, Oval, Text Box, WordArt, Fill Color, Line Color, Shadow, 3-D** etc. All the functions below map to those icons.

1) Insert & Place a Shape

Purpose: Add visual elements (boxes, circles, arrows) for diagrams or callouts.

Steps:

1. Go to **Insert** → **Shapes**.
2. Choose a shape (e.g., **Rectangle**).
3. **Click-drag** on the sheet to draw it.
 - **Tip:** Hold **Shift** for a perfect square/circle.
 - **Tip:** Hold **Alt** while dragging to snap edges to cell boundaries.

2) Add Text to a Shape

Purpose: Label your shapes (e.g., “Start”, “Total Sales”).

Steps:

1. Select the shape, start typing your text.
2. Use **Home** → **Font/Alignment** or **Shape Format** → **Text** to style it.
 - **Wrap text in shape:** Right-click shape → **Format Shape** → **Text Options** → **Text Box** → **Wrap text in shape**.

3) Text Box (independent text anywhere)

Purpose: Place headings/notes not tied to a cell.

Steps:

1. **Insert** → **Text Box** (or **Insert** → **Text** → **Text Box**).
2. Click and type. Format using **Home** or **Shape Format**.

4) WordArt (decorative headings)

Purpose: Stylish titles with outline, shadow, gradient.

Steps:

1. **Insert** → **WordArt**, pick a style.
2. Type your text.
3. Adjust via **Shape Format** → **WordArt Styles**.

5) Lines, Arrows & Connectors

Purpose: Show relationships/flow.

Steps (straight line/arrow):

1. **Insert** → **Shapes** → **Lines/Arrow**.
2. Click-drag to draw.
 - **Tip:** Hold **Shift** to constrain to 45°/90°.

Steps (connectors that “stick” to shapes):

1. **Insert** → **Shapes** → **Elbow/Curved/Straight Connector**.
2. Hover near a shape’s edge until **connection dots** appear, then click to attach.
3. Drag the other end to the next shape until it snaps.
 - Connectors **stay attached** when you move shapes.

6) Quick Styling (Shape Fill, Outline, Effects)

Purpose: Make visuals clear and consistent.

Steps:

1. Select the shape → **Shape Format** tab.
2. Use:
 - **Shape Fill:** solid, gradient, picture, or pattern fills.
 - **Shape Outline:** color, **Weight** (thickness), **Dashes**.
 - **Shape Effects:** **Shadow**, **Glow**, **Soft Edges**, **Bevel**, **3-D Rotation**.
3. For advanced control: Right-click → **Format Shape** to open the side pane (fine-tune transparency, gradient stops, shadow angle, etc.).

7) Size, Position & Rotation

Purpose: Exact dimensions and orientation.

Steps:

1. Select shape → **Shape Format** → **Size** (set **Height/Width** numerically).
2. **Rotate:** **Shape Format** → **Arrange** → **Rotate** (Rotate 90°, Flip, or **More Rotation Options**).
3. Keep proportions: Right-click → **Size and Properties** → **Size** → **Lock aspect ratio**.

8) Align, Distribute & Snap

Purpose: Tidy, professional layout.

Steps (Align):

1. Select **2+ shapes** (Ctrl+click each).
2. **Shape Format** → **Align** → **Align Left/Center/Right/Top/Middle/Bottom**.

Steps (Distribute):

1. With 3+ items selected → **Align** → **Distribute Horizontally/Vertically**.

Snap options:

- **Align** → **Snap to Grid** or **Snap to Shape** for neat edges.

10) Group / Ungroup

Purpose: Move/format multiple shapes as one object.

Steps:

1. Select all target shapes (Ctrl+click).
2. **Shape Format** → **Group** → **Group**.
3. To change one part later: **Group** → **Ungroup** (or **Regroup**).

11) Edit Shape / Edit Points

Purpose: Customize geometry.

Steps:

1. Select shape → **Shape Format** → **Edit Shape** → **Edit Points**.
2. Drag points/handles to tweak corners and curves.
3. **Edit Shape** → **Change Shape** to swap rectangle → rounded rectangle, etc., without redrawing.

13) Merge Shapes (if available)

Purpose: Build custom silhouettes: **Union, Combine, Fragment, Intersect, Subtract**.

Steps:

1. Select 2+ shapes → **Shape Format** → **Merge Shapes** → choose operation.
 - If you don't see it, customize Ribbon or ensure a recent Excel version.

Drawing toolbar

What it is:

A set of tools inside Excel to add shapes, text boxes, lines, WordArt and other visual elements to a worksheet. Useful for flowcharts, callouts, highlights, and making reports look professional.

How to open it:

- Excel Ribbon → **Insert** → look under **Illustrations** → choose **Shapes / Text Box / Icons / SmartArt**. (Older Excel versions had a separate "Drawing" toolbar.)

Common tools & uses:

- **Shapes (rectangle, oval, arrow, callout):** draw boxes and arrows to annotate tables or make diagrams.
- **Text Box:** place text anywhere independent of cells.
- **WordArt:** decorative headings.
- **Lines & Connectors:** connect shapes (useful for flowcharts).
- **Format Shape pane:** change fill, outline, effects (shadow, glow), size, rotation.

Step-by-step example: Add a labeled arrow pointing to a total cell.

1. Insert → Shapes → choose Arrow. Draw it pointing to the cell.
2. Insert → Text Box → type "Grand Total" and place near arrow.
3. Select arrow → right-click → Format Shape → adjust thickness and arrowhead.
4. Select both → Home → Arrange to align.

Tips & best practice:

- Use **Align** and **Distribute** (Format → Align) to keep shapes neat.
- Group objects (select multiple → right-click → Group) so they move together.
- Use **Send to Back / Bring to Front** to layer shapes over cells or images.
- Avoid placing important text solely inside shapes — keep data in cells so formulas and sorting still work.

Common mistakes:

- Covering cells with shapes making cell contents hard to select — right-click shape → Send to Back or set shape to “No fill”.
- Not grouping related shapes, causing misalignment when resizing.

Exercise: Create a mini flowchart showing steps: Input → Process → Output. Use 3 shapes and arrows.

Freeze Panes

What it is:

Keeps selected rows and/or columns visible while you scroll through the rest of the worksheet. Vital for large datasets — keeps headers or key columns in view.

When to use:

Large tables where you want headings (top row) or ID columns (first column) to remain visible while scrolling.

How to use (steps):

- To freeze top row: View → Freeze Panes → **Freeze Top Row**.
- To freeze first column: View → Freeze Panes → **Freeze First Column**.
- To freeze both rows & columns (e.g., top 1 row and left 1 column) place cursor in cell B2, then View → Freeze Panes → **Freeze Panes**. Excel freezes all rows above and columns left of the active cell.

Check it worked: you'll see a thin line showing the frozen boundary. Scroll — frozen rows/columns stay visible.

Tips & shortcuts:

- Shortcut to unfreeze: View → Freeze Panes → **Unfreeze Panes**.
- Use Freeze Panes instead of splitting if you want only headers to be fixed.

Common mistakes:

- Placing the active cell in the wrong place leads to freezing the wrong rows/columns. Remember: Excel freezes *above* and *left* of the active cell.

Exercise: Freeze top row and first column on a student marks sheet.

Splitting the worksheet

What it is:

Divides the worksheet window into two or four panes that can scroll independently. Useful to compare distant parts of the same sheet simultaneously.

How to split:

1. Click the cell where you want split lines to appear (cell determines vertical/horizontal splits).
2. View → **Split**. Excel creates split bars. You can drag the split bars to adjust.
3. To remove split: View → Split (toggle off).

When to use vs Freeze Panes:

- **Split** → compare separate areas of the same worksheet (each pane scrolls independently).
- **Freeze** → keep headings fixed while scrolling (pane does not scroll independently).

Practical example: Compare totals at top of sheet with details far below. Use split to put top section in one pane and details in another.

Tips:

- You can scroll each pane separately; useful for cross-checking rows.
- When using split with multiple monitors, maximize window to get better view.

Exercise: Split a large sales sheet horizontally so top 20 rows remain in the top pane and you can scroll the rest below.

Goal Seek

What it is:

A simple what-if tool that finds the input value required to reach a specific output. Excel changes one cell (input) until the formula cell (output) equals your target value.

When to use:

Solve single-variable problems like “What monthly payment achieves a target loan payment?” or “How many units must we sell to reach ₹50,000 revenue?”

How to use (example): Suppose A1 contains price, A2 contains quantity, A3 has formula =A1*A2 (total revenue). You want total revenue = ₹50,000 by changing quantity (A2).

1. Data → What-If Analysis → **Goal Seek**.
2. Set cell: A3 (the formula/result cell).
3. To value: 50000.

4. By changing cell: A2.
5. Click **OK**. Excel iterates and returns the result.

Notes:

- Goal Seek changes only one variable.
- The cell you change must be directly or indirectly referenced by the target formula.

Limitations:

- Only one input variable at a time.
- Nonlinear or multi-solution problems may not behave as expected.
- Goal Seek uses iteration — sometimes it may not converge if the relationship is complex.

Tips:

- For more than one variable, use **Solver** (add-in).
- Make initial guess sensible — helps convergence.

Exercise: Create a loan payment example: PV in A1, rate in A2, period in A3, payment formula in A4 ($\text{=PMT}(A2/12,A3,-A1)$), then use Goal Seek to find period needed for a monthly payment of ₹5,000.

Multiple Workbooks

What it is:

Working with more than one Excel file at the same time. Useful for splitting data by year, department, or file type while linking or comparing between them.

Common tasks:

- **Open multiple workbooks:** File → Open (or drag files).
- **View side-by-side:** View → **View Side by Side** to compare two workbooks; use **Synchronous Scrolling** to scroll both at once.
- **Link between workbooks:** Use formulas referencing other workbooks.

How to link (external reference) example:

- In Book1.xlsx cell A1 you want value from Book2.xlsx Sheet1 cell B2:
 $\text{=[Book2.xlsx]Sheet1!B2}$
- If Book2 is closed, the formula still works; Excel shows the full path:
 $\text{='C:\Folder\[Book2.xlsx]Sheet1'!B2}$

Updating links:

- When opening a workbook with external links, Excel prompts to update links. Click **Update** to pull latest values.

- Manage Links: Data → **Edit Links** (shows source files and options: Update, Change Source, Break Link).

Copying data vs linking:

- Copy-paste: snapshot (no live update).
- Linking: dynamic (updates when source changes).

Practical examples:

- Consolidate monthly files into a yearly summary using links.
- Departmental budgets stored in separate files and linked to a master workbook.

Tips & best practice:

- Keep file paths short and stable; moving source files breaks links.
- Use named ranges in source workbook for clearer links: `='[Book2.xlsx]Sheet1'!Sales_Q1`
- If sharing, ensure recipients have access to source files or break links and paste values.

Common issues & troubleshooting:

- Broken link error → use Edit Links → Change Source or update path.
- Performance slowdown when many external links exist.
- Circular references if two workbooks reference each other.

Exercise: Create two small workbooks, enter numbers in one, and reference them in the other. Change the source and reopen to see updates.

Pivot Table and Pivot Chart

What they are:

- **Pivot Table:** A powerful tool to summarize, analyze, explore, and present large amounts of data quickly (no formulas required).
- **Pivot Chart:** A visual chart built from a Pivot Table that updates when the Pivot Table changes.

When to use:

Summarize sales by region, count student results by grade, analyze trends by month — anytime you need flexible summary and reporting.

How to create a Pivot Table (step-by-step):

1. Ensure your data is in a proper table format: header row with unique field names, no blank rows/columns.
2. Select any cell in the data → **Insert** → **PivotTable**.
3. Choose where to place it (New Worksheet recommended) → OK.

4. PivotTable Field List appears. Drag fields to:

- **Rows:** categories you want as rows (e.g., Region, Product).
- **Columns:** categories across top (e.g., Quarter).
- **Values:** numeric data to summarize (e.g., Sales, Quantity). Default is SUM — can change to COUNT, AVERAGE.
- **Filters:** top-level filter (e.g., Year or Product Type).

Key features:

- **Group data:** Right-click a date field → Group (by month, quarter, year). For numeric fields: group into ranges.
- **Value Field Settings:** choose Sum/Count/Average/Max/Min; show values as % of row/column/total.
- **Refresh data:** Right-click PivotTable → **Refresh** when source data changes.
- **Slicers & Timelines:** visual filters — Insert → Slicer or Timeline (for date fields). Make interactive dashboards.
- **Calculated Field:** Create custom calculation inside PivotTable: PivotTable Analyze → Fields, Items & Sets → Calculated Field.

Pivot Chart:

- With PivotTable selected → PivotTable Analyze → **PivotChart** → choose chart type.
- Chart is linked — changes in PivotTable (filters, slicers) update the chart.

Practical classroom example:

Student marks dataset with columns: Student, Class, Subject, Marks, Gender. Use PivotTable to compute:

- Average marks per subject.
- Count of students per class.
- Distribution of grades by gender (use grouping to create grade ranges).

Performance & limits:

- PivotTables can handle large datasets, but very big datasets may be slower.
- For massive data, use Data Model / Power Pivot (advanced).

Tips & best practices:

- Convert raw data into an **Excel Table** first (Insert → Table). PivotTables will auto-expand as you add rows.
- Name your PivotTable sheets clearly for sharing.
- Use slicers for interactive filtering — great in classroom demos.

- Keep a copy of raw data; avoid changing source columns headings after creating the pivot.

Common mistakes:

- Blank rows/columns in source data break field detection.
- Forgetting to Refresh → shows stale results.

Exercise: Provide a sales dataset (Date, Region, Salesperson, Amount). Ask students to create a PivotTable to show total sales by Region and a PivotChart (column chart) showing sales by month using Group by Date.

Macros (and basic VBA)

What macros are:

Macros automate repetitive tasks by recording actions or by writing small programs in VBA (Visual Basic for Applications).

Record a macro (basic steps):

1. View → Macros → **Record Macro** (or Developer tab → Record Macro).
2. Give it a name (no spaces), optional shortcut key, and choose where to store (This Workbook or Personal Macro Workbook).
3. Perform the actions you want recorded (formatting, inserting formulas, sorting).
4. Stop Recording (View → Macros → Stop Recording).

Run a macro:

- View → Macros → **View Macros** → select → Run.
- Or use the assigned shortcut, or assign macro to a button/form control: Developer → Insert → Button → assign macro.

Edit a macro (VBA):

- View → Macros → View Macros → select → Edit → opens the VBA editor where you can see and modify code.
- Basic knowledge of VBA helps to make macros robust (error handling, loops).

Storing macros:

- **This Workbook:** macros available only in the workbook.
- **Personal Macro Workbook (PERSONAL.XLSB):** macros available in all Excel sessions on that PC.

Macro security:

- Excel warns on opening workbooks with macros. Set macro security: File → Options → Trust Center → Trust Center Settings → Macro Settings.
- Only enable macros from trusted sources.

1. Lookup and Reference Functions in Excel

Meaning:

Lookup and Reference functions are used to **search for specific data** in a worksheet and **return related information** from another cell or range.

These functions help you quickly find data from large tables.

Common Lookup and Reference Functions:

1. VLOOKUP (Vertical Lookup):

- **Purpose:** Searches for a value in the first column of a table and returns a value from the same row in another column.
- **Syntax:**
- =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
- **Example:**
If you have a list of students with their marks:
- =VLOOKUP("Ravi", A2:C10, 3, FALSE)

→ This will find "Ravi" in column A and return his marks from the 3rd column.

2. HLOOKUP (Horizontal Lookup):

- **Purpose:** Searches for a value in the **top row** of a table and returns data from the same column in a row below.
- **Syntax:**
- =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
- **Example:**
- =HLOOKUP("Total", A1:F5, 3, FALSE)

→ Finds "Total" in the top row and returns the value from the 3rd row of that column.

3. MATCH Function:

- **Purpose:** Finds the **position** of a value in a range.
- **Syntax:**
- =MATCH(lookup_value, lookup_array, [match_type])

- **Example:**
- =MATCH("Ravi", A2:A10, 0)

→ Returns the position of "Ravi" in that list.

✓ 4. INDEX Function:

- **Purpose:** Returns the value of a cell in a specific row and column in a given range.
- **Syntax:**
- =INDEX(array, row_num, [column_num])
- **Example:**
- =INDEX(B2:D10, 3, 2)

→ Returns the value from the 3rd row and 2nd column of the given range.

🧠 Key Point for Exam:

- VLOOKUP → Vertical Search
- HLOOKUP → Horizontal Search
- MATCH → Finds Position
- INDEX → Returns Value

✓ 2. Data Validation in Excel

📘 Meaning:

Data Validation is used to **control the type of data or value** that a user can enter in a cell. It helps to **reduce errors** in data entry and maintain accuracy.

📌 Steps to Apply Data Validation:

1. Select the **cell(s)** where you want to apply validation.
2. Go to **Data tab** → **Data Validation** → **Data Validation**.
3. In the dialog box, choose the **type of data** you want to allow.
4. Set the conditions (like range, list, text length, etc.).

5. Click **OK**.

Types of Data Validation:

Type	Description	Example
Whole Number	Allows only whole numbers	Allow marks between 0 and 100
Decimal	Allows decimal values	Allow values between 1.0 and 10.0
List	User selects from a dropdown list	Allow city names: "Ahmedabad, Surat, Rajkot"
Date	Allows only date values	Between 01-01-2024 and 31-12-2024
Time	Allows only time values	Between 09:00 and 17:00
Text Length	Restricts number of characters	Allow only 10 characters for ID
Custom	Allows formula-based validation	=ISNUMBER(A1) to allow only numbers

Example:

If you want only marks between **0 and 100**:

1. Select the marks column.
2. Go to **Data** → **Data Validation**.
3. Choose **Whole Number** → Between → 0 and 100 → OK.

Now, Excel will not accept values outside this range.

Defining Valid Sets of Values for Ranges of Cells using Data Validation

Meaning:

This feature allows you to create a list of acceptable (valid) values for one or more cells. It means users can only enter data that matches your predefined list — helping you avoid wrong entries.

Example:

Let's say you have a column for selecting a Department (like "IT", "HR", "Finance", "Sales"). You can define these values using Data Validation (List option) so that users can select only from the list.

Steps to Define Valid Sets of Values:

1. Select the range of cells where you want to apply validation.

2. Click on the Data tab → Data Validation.
3. In the Allow box, choose List.
4. In the Source box, type your valid values separated by commas.
5. IT, HR, Finance, Sales
6. Click OK.

Now, a drop-down arrow appears in those cells — users can select only from those options.

Example 2 (Using a Range):

If you have a list of departments already written in cells A1:A4, you can set the Source as:

=A\$1:\$A\$4

This makes your validation dynamic — if you change the list later, it automatically updates.

Advantages:

- Prevents typing errors
- Ensures uniform data entry
- Saves time by providing ready-made choices

Key Points for Exam:

- Use Data Validation → List to define valid sets of values.
- Can use typed values or a cell range as source.
- Helps maintain data accuracy and consistency.

2. Reordering and Summarizing Data – Sorting Worksheet Data

Meaning:

Sorting means arranging data in a specific order — either alphabetically, numerically, or by date. It helps to quickly analyze or summarize the worksheet data.

Types of Sorting:

Type	Description	Example
Ascending Order	Smallest to Largest / A to Z / Oldest to Newest	Roll Numbers: 1, 2, 3, 4
Descending Order	Largest to Smallest / Z to A / Newest to Oldest	Marks: 95, 90, 85, 80

Steps to Sort Data:

1. Select the data range you want to sort.
2. Go to the Data tab → Sort & Filter group.
3. Click on:
 - Sort A to Z → Ascending order
 - Sort Z to A → Descending order
4. Or choose Custom Sort for advanced options (e.g., sorting by multiple columns).

Example:

You have a list of students with names and marks.

Name	Marks
Ravi	70
Meena	85
Ajay	60

If you sort by Marks (Descending), the result will be:

Name	Marks
Meena	85
Ravi	70
Ajay	60

Custom Sorting Example:

If you have multiple columns like Department and Employee Name, you can first sort by Department, and then by Name.

Advantages of Sorting:

- Makes data easy to read and compare
- Helps in summarizing results
- Saves time in data analysis



Excel Functions

Excel provides many **built-in functions** to perform calculations quickly and easily.

Each function is a **ready-made formula** that helps to do specific types of work — such as finding totals, averages, dates, or financial values.



1. Date and Time Functions



Meaning:

Date and Time functions are used to **display, calculate, or extract** information from dates and times.

Function	Description	Example
TODAY()	Displays the current date	=TODAY() → 23-Oct-2025
NOW()	Displays the current date and time	=NOW() → 23-Oct-2025 09:30 AM
DATE(year, month, day)	Creates a date from given numbers	=DATE(2025,10,23) → 23-Oct-2025
DAY(date)	Returns the day from a date	=DAY("23-Oct-2025") → 23
MONTH(date)	Returns the month number	=MONTH("23-Oct-2025") → 10
YEAR(date)	Returns the year	=YEAR("23-Oct-2025") → 2025
HOUR(time)	Returns hour value from time	=HOUR("09:30 AM") → 9
MINUTE(time)	Returns minutes from time	=MINUTE("09:30 AM") → 30



Use: To calculate age, duration, or project deadlines.



2. Statistical Functions



Meaning:

Statistical functions are used to **analyze data** and find values like average, minimum, maximum, or count.

Function	Description	Example
AVERAGE(range)	Calculates average of numbers	=AVERAGE(A1:A5)
MAX(range)	Finds the highest value	=MAX(A1:A5)
MIN(range)	Finds the lowest value	=MIN(A1:A5)
COUNT(range)	Counts how many cells have numbers	=COUNT(A1:A10)
COUNTA(range)	Counts non-empty cells	=COUNTA(A1:A10)
MEDIAN(range)	Finds the middle value	=MEDIAN(A1:A5)
STDEV(range)	Calculates standard deviation	=STDEV(A1:A10)



Use: Helpful for student marks analysis, surveys, and reports.

3. Mathematical Functions

Meaning:

Mathematical functions help to **perform calculations** like addition, subtraction, rounding, and power.

Function	Description	Example
SUM(range)	Adds up numbers	=SUM(A1:A10)
PRODUCT(range)	Multiplies numbers	=PRODUCT(A1:A3)
SQRT(number)	Finds square root	=SQRT(25) → 5
POWER(number, power)	Raises to a power	=POWER(2,3) → 8
ROUND(number, num_digits)	Rounds a number to given decimals	=ROUND(12.345, 2) → 12.35
ABS(number)	Returns absolute (positive) value	=ABS(-10) → 10
INT(number)	Removes decimals and shows integer part	=INT(12.67) → 12

 **Use:** To calculate totals, profits, percentages, and mathematical results.

4. Financial Functions

Meaning:

Financial functions are used for **money-related calculations**, like interest, payments, and investments.

Function	Description	Example
PMT(rate, nper, pv)	Calculates loan payment	=PMT(10%/12, 12, 10000) → -879.16
FV(rate, nper, pmt)	Finds future value of investment	=FV(10%/12, 12, -1000)
PV(rate, nper, pmt)	Finds present value of investment	=PV(10%/12, 12, -1000)
RATE(nper, pmt, pv, fv)	Finds interest rate	=RATE(12, -1000, 10000)
NPER(rate, pmt, pv)	Finds number of periods	=NPER(10%/12, -1000, 10000)

 **Use:** To calculate EMI, interest, savings growth, or investments.

5. Database Functions

Meaning:

Database functions work with **structured data** (like tables with headers). They perform calculations only on records that **match certain conditions**.

Function	Description	Example
DSUM(database, field, criteria)	Adds numbers that meet condition	=DSUM(A1:C10, "Marks", E1:E2)
DAVERAGE(database, field, criteria)	Finds average of matching records	=DAVERAGE(A1:C10, "Marks", E1:E2)
DCOUNT(database, field, criteria)	Counts matching records	=DCOUNT(A1:C10, "Marks", E1:E2)
DMAX(database, field, criteria)	Finds maximum matching value	=DMAX(A1:C10, "Marks", E1:E2)
DMIN(database, field, criteria)	Finds minimum matching value	=DMIN(A1:C10, "Marks", E1:E2)

 **Use:** To perform conditional calculations in a table (e.g., total marks of students from “IT” department).

Key Points for Exam:

- **Date-Time:** Works with date and time values (e.g., TODAY, NOW, YEAR).
- **Statistical:** Analyzes data (e.g., AVERAGE, MAX, MIN).
- **Mathematical:** Performs numeric calculations (e.g., SUM, ROUND, POWER).
- **Financial:** Deals with money and investments (e.g., PMT, FV).
- **Database:** Works with filtered or condition-based data (e.g., DSUM, DCOUNT).